Kotha Pratheek Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.78
Pallavi Model School	Class XII (CBSE)	2024	95.40%
FIITJEE World School	Class X (SSC)	2022	10.00%

RELEVANT COURSES

Systems Related : Operating Systems 1 , Database Management Systems 1.

Theoretical Computer Science : Data Structures and Algorithms , Discrete Mathematics.

Artificial Intelligence : Foundations of Machine Learning , Deep Learning , AI for Humanity .

PROJECTS

e-Yantra Robotics Competition (eYRC 2025-26) | IIT Bombay | Source Code:

- Developed a ROS-based multi-robot warehouse automation system where three holonomic robots use overhead ArUco marker tracking to autonomously sort and transport color-coded crates from four pickup zones to designated drop zones, Securing **3rd** out of the 449 participating teams in the theme Holo Battalion.
- Built the ROS communication framework and MQTT-based interface to enable reliable, low-latency coordination between the vision system, navigation stack, and embedded robot controllers.
- Implemented low-level Arduino (C++) firmware and designed/tuned PID trajectory tracking controllers for precise closed-loop motion control, motor actuation, and sensor integration.

Foundations of Machine Learning | Course Projects | Link :

- Built a from-scratch backpropagation/autograd engine (Value class with computation graph construction and gradient flow), along with modular NN components (Linear, ReLU, Sigmoid, Sequential), loss functions, and an SGD optimizer.
- Built a mini machine learning library similar to scikit-learn, with tools for loading data, preprocessing, splitting data for testing, and classical models like Logistic Regression and LDA

Multimodal PDF based RAG System for Joint Text and Image Embedding-Based Search | Link :

- Built a multimodal RAG pipeline that extracts and retrieves PDF text and embedded images using BGE and CLIP embeddings with FAISS.
- Implemented independent text and image retrieval with similarity-based image filtering to provide relevant visual context only when appropriate.
- Integrated retrieved text and images with a multimodal LLM to generate context-grounded answers with source citations.

SKILLS

Languages : C , C++ , Python , PostgreSQL

Libraries & Frameworks : Numpy , Pandas , Matplotlib , scikit-learn , ROS2 , PyTorch , OpenCV , paho-mqtt.

Tools : Git

POSITIONS OF RESPONSIBILITY

Matrix Theory - Undergraduate Teaching Assistant under Dr. GVV Sharma.

Work Done: Helped students with problem solving, conducted vivas and evaluations.

Probability and Random Variables - Teaching Assistant under Dr. Anjana Mahesh

Work Done: Evaluated answer scripts, held tutorial sessions with problem-solving support.

Robotix IITH :

- Core Member, Programming & Machine Learning Domain, Robotix IITH
- Represented IIT Hyderabad at the e-Yantra Robotics Competition (2025-26), securing 3rd place in the finals held at IIT Bombay.

SCHOLASTIC ACHIEVEMENTS

- **National Standard Examination in Physics - 2023 -** State Top 1%.
- **National Standard Examination in Chemistry - 2023 -** Center Top 10%.

REASEARCH EXPERIENCE

Machine Unlearning Reasearch:

Working under Dr. Konda Reddy Mopuri

- Exploring state-of-the-art machine unlearning methods through a comprehensive study of recent research literature.
- Investigating retain-set-free machine unlearning techniques, focusing on efficient unlearning without access to the retain dataset and evaluating their utility-forgetting trade-offs.